

Projection Theorem

Vector Space

Inner Product

- 1. **Bilinearity:** $\langle ax + by, z \rangle = a\langle x, z \rangle + b\langle y, z \rangle$
- 2. **Symmetry:** $\langle x, y \rangle = \langle y, x \rangle$
- 3. **Positivity:** $\langle x, x \rangle \geq 0$, and $\langle x, x \rangle = 0$ iff $x = 0$

Example: $\langle X, Y \rangle = \frac{1}{N} \sum_{i=1}^N X_i Y_i$, $\langle X, Y \rangle = \mathbb{E}[XY]$

Ineq $\|x + y\| \leq \|x\| + \|y\|$, Cauchy Schwarz $|\langle x_1, x_2 \rangle| \leq \|x_1\| \|x_2\|$

Pythagorean Theorem: $\langle x_1, x_2 \rangle = 0 \Rightarrow \|x_1 + x_2\|^2 = \|x_1\|^2 + \|x_2\|^2$.

Projection Theorem

(1) $\hat{Y} = \Pi(Y|\mathcal{L})$ is unique (2) $\langle Y - \hat{Y}, \tilde{Y} \rangle = 0$ for all $\tilde{Y} \in \mathcal{L}$ or $Y - \Pi(Y|\mathcal{L}) \perp \tilde{Y}$ for all $\tilde{Y} \in \mathcal{L}$. $\Leftrightarrow$ Projection.

Linear Regression

$\beta_0 = \mathbb{E}[X X^\top]^{-1} \mathbb{E}[XY]$, $\pi(Y|\mathcal{L}) = \mathbb{E}[Y|X] = X^\top \beta_0$

Decision Theory

$\{\mathcal{P}_\theta : \theta \in \Theta\}$, let Z be a random variable with distribution $\mathcal{P}_\theta$ for some $\theta \in \Theta$, with $\mathcal{Z}$ as the sample space.
Loss Function: $L(\theta, d)$ real-valued on $A \times \Theta$
Decision Rule: $d(Z)$ $d : \mathcal{Z} \rightarrow A$ Risk : $R(\theta, d) = \mathbb{E}_\theta[L(\theta, d(Z))]$
Expected Loss: $R(\theta, d) = \mathbb{E}_\theta[(\theta - d(Z))^2] = (\theta - \mathbb{E}_\theta[d(Z)])^2 + \text{Var}_\theta(d(Z))$

Prior: $\pi(\theta) = \begin{cases} 1, & \text{if } 0 \leq \theta \leq 1 \\ 0, & \text{otherwise} \end{cases}$

Likelihood: $l(Z|\theta) = \prod_{i=1}^N \theta^{Z_i} (1 - \theta)^{1-Z_i} = \theta^{S_N} (1 - \theta)^{N-S_N}$

Posterior: $\pi(\theta|Z) = \frac{l(Z|\theta) \cdot \pi(\theta)}{\int_0^1 l(Z|t) \pi(t) dt} = \frac{\theta^{S_N} (1-\theta)^{N-S_N}}{\int_0^1 t^{S_N} (1-t)^{N-S_N} dt}$

$\beta(\alpha_1, \alpha_2) = \int_0^1 t^{\alpha_1-1} (1-t)^{\alpha_2-1} dt$.
 $\Rightarrow$ Posterior distribution: $\theta|\mathbf{Z} = \mathbf{z} \sim \text{Beta}(S_N + 1, N - S_N + 1)$

median of a $\beta(\alpha_1, \alpha_2)$ distributed random variable is $\frac{\alpha_1 - \frac{1}{3}}{\alpha_1 + \alpha_2 - \frac{2}{3}}$

Properties of posterior:

$\mathbb{E}[\theta|\mathbf{Z} = \mathbf{z}] = \frac{S_N + 1}{N + 2}$, $\text{Var}(\theta|\mathbf{Z} = \mathbf{z}) = \frac{(S_N + 1)(N - S_N + 1)}{(N + 2)^2 (N + 3)}$

Finding a Decision Rule or "Best Guess" for θ
 $\theta \sim \text{Beta}(\alpha_1, \alpha_2)$, $\theta|\mathbf{Z} = \mathbf{z} \sim \text{Beta}(S_N + \alpha_1, N - S_N + \alpha_2)$
Laplace-Bayes $\theta \sim U[0, 1] = \text{Beta}(1, 1)$

Least-Favorable-Bayes $\theta \sim \text{Beta}(\alpha_1, \alpha_2)$, $\alpha_1 = \alpha_2 = \sqrt{\frac{N}{4}}$,

$d_{LFB}(\mathbf{Z}) = \frac{S_N + \frac{1}{2}\sqrt{N}}{N + \sqrt{N}}$, $R(\theta, d_{LFB}) = \frac{N}{4(N + \sqrt{N})^2}$

summary Under a general prior $\theta \sim \text{Beta}(\alpha_1, \alpha_2)$, the posterior mean decision rule is as follows:

$d_{BB}(\mathbf{Z}) = \frac{S_N + \alpha_1}{N + \alpha_1 + \alpha_2} = \frac{N}{N + \alpha_1 + \alpha_2} (\underbrace{\frac{S_N}{N}}_{\text{sample mean}}) + \underbrace{\frac{\alpha_1 + \alpha_2}{N + \alpha_1 + \alpha_2}}_{\text{prior mean}} (\frac{\alpha_1}{\alpha_1 + \alpha_2})$

Complete Class Theorem: Any admissible estimator is a Bayes rule (or a limit of Bayes rule) with respect to some prior.
Admissibility + constant risk $\Rightarrow$ Minimax: $\inf_{d \in A} \sup_{\theta \in \Theta} R(\theta, d)$.

Average Risk (Bayes Risk): $L(\theta, a) = (\theta - a)^2$,
 $R(\theta, d) = \mathbb{E}(L(\theta, d(\mathbf{Z})) | \theta) = \int L(\theta, d(\mathbf{Z})) f(\mathbf{z} | \theta) d_m(\mathbf{z})$
 $r(\pi, d) = \int R(\theta, d) \pi(\theta) d_m(\theta) = \int (\int L(\theta, d(\mathbf{Z})) f(\mathbf{z} | \theta) \pi(\theta) d_m(\theta)) d_m(\mathbf{z})$

$\pi(\theta | \mathbf{z}) \propto f(\mathbf{z} | \theta) \cdot \pi(\theta) \Rightarrow d_\pi(\mathbf{z}) = \arg \min_{a \in A} \int L(\theta, a) \pi(\theta | \mathbf{z}) d_m(\theta)$

$d_\pi(\mathbf{z}) = \arg \min_{a \in A} \mathbb{E}[(\theta - a)^2 | \mathbf{Z} = \mathbf{z}]$
 $= \arg \min_{a \in A} \mathbb{E}[(\theta - \mathbb{E}(\theta | \mathbf{Z} = \mathbf{z}) - (a - \mathbb{E}(\theta | \mathbf{Z} = \mathbf{z})))^2 | \mathbf{Z} = \mathbf{z}]$
 $= \arg \min_{a \in A} \{ (a - \mathbb{E}(\theta | \mathbf{Z} = \mathbf{z}))^2 + \text{Var}(\theta | \mathbf{Z} = \mathbf{z}) \}$

Thus, $d_\pi^*(\mathbf{z}) = \mathbb{E}(\theta | \mathbf{Z} = \mathbf{z})$ under squared error loss.

summary Bayes risk allows us to rank decision rules (since it is independent of the true value of θ .), the ranking depends on our prior.

K-normal Means & James-Stein Estimator

Settings: $Y = (Y_1, \dots, Y_N)'$ and $X = (\underbrace{X_1, \dots, X_N}_{K \times 1})'$

Let $m := \mathbb{E}(Y) = \mathbb{E}(Y | X)$, $\hat{m}_i = \hat{m}(X_i)$ is our prediction.

Loss function: $\mathbb{E}(\|\hat{m} - m\|^2) = \sum_{i=1}^N \mathbb{E}((\hat{m}(X_i) - m(X_i))^2)$
 $Y_i = m(x_i) + \sigma U_i$, $U_i | X_i \sim N(0, 1)$, $U_i | X_i \sim N(0, 1)$

Apparent Error

$\mathbb{E}[\|\mathbf{Y} - \hat{\mathbf{m}}\|^2] = \mathbb{E}[\|\mathbf{Y} - \mathbf{m}\|^2] + \mathbb{E}[\|\mathbf{m} - \hat{\mathbf{m}}\|^2] + 2\mathbb{E}[(\mathbf{Y} - \mathbf{m}) \cdot (\mathbf{m} - \hat{\mathbf{m}})]$
 $\mathbb{E}[\|\mathbf{Y} - \mathbf{m}\|^2] = N\sigma^2$

$2\mathbb{E}[(\mathbf{Y} - \mathbf{m})'(\mathbf{m} - \hat{\mathbf{m}})] = 2\mathbb{E}[(\mathbf{Y} - \mathbf{m})' \mathbf{m}] - 2\mathbb{E}[(\mathbf{Y} - \mathbf{m})' \hat{\mathbf{m}}]$
 $= -2\mathbb{E}\left[\sum_{i=1}^N (Y_i - m_i) \hat{m}_i\right] = -2 \sum_{i=1}^N \text{Cov}(Y_i, \hat{m}_i)$

Defining degrees of freedom as $df(\hat{\mathbf{m}}) = \sum_{i=1}^N \frac{\text{Cov}(Y_i, \hat{m}_i)}{\sigma^2} \Rightarrow$
 $\mathbb{E}[\|\mathbf{m} - \hat{\mathbf{m}}\|^2] = \underbrace{-N\sigma^2}_{\text{Risk / MSE}} + \underbrace{\mathbb{E}[\|\mathbf{Y} - \mathbf{m}\|^2]}_{\text{Noise Floor}} + \underbrace{2\sigma^2 df(\hat{\mathbf{m}})}_{\substack{\text{Model Complexity /} \\ \text{Degrees of Freedom}}}$

K-normal Means: Assuming $m(x) = \sum_{k=1}^K \alpha_k g_k(x)$
Gram-Schmidt Orthogonalization:

Set up: $\langle f, g \rangle = \frac{1}{N} \sum_{i=1}^N f(x_i) g(x_i)$, $\|f\| = \sqrt{\langle f, f \rangle} = [\frac{1}{N} \sum_{i=1}^N f(x_i)^2]^{\frac{1}{2}}$
Given basis functions $g_1(x), \dots, g_K(x)$, the recursive generation of ϕ is
 $f_K(x) = g_K(x) - \sum_{k=1}^{K-1} \frac{\langle g_K, f_k \rangle}{\langle f_k, f_k \rangle} f_k(x)$, $\phi_k(x) = \frac{f_1(x)}{\|f_k\|}$.
Properties: $\langle \phi_k, \phi_j \rangle = 0 \wedge \|\phi_k\| = 1 \quad \forall j, k \text{ s.t. } k \neq j$

MLE of θ :
Set $m(x_i) = \sum_{k=1}^K \theta_k \phi_k(x_i)$, $\omega(x) = (\phi_1(x), \dots, \phi_K(x))'$
 $\hat{\theta}_{ML}$ by finding the log likelihood is
 $l(\mathbf{Y}|\mathbf{X}; \theta) = \frac{-N}{2} \ln(2\pi) - N \ln(\sigma) - \frac{1}{2\sigma^2} \sum_{i=1}^N (Y_i - \omega(X_i)' \theta)^2 \Rightarrow$
 $\hat{\theta}_{ML} = [\frac{1}{N} \sum_{i=1}^N \omega(X_i) \omega(X_i)'] \times [\frac{1}{N} \sum_{i=1}^N \omega(X_i) Y_i]$
Given $\frac{1}{N} \sum_{i=1}^N \omega(X_i) \omega(X_i)' = I_k$, $\hat{\theta}_{ML} = \mathbf{Z} = \frac{1}{N} \sum_{i=1}^m W_i Y_i$, $W_i = \omega(X_i)$.

The estimation is $\hat{m}(x) = \omega(x)' \hat{\theta}_{ML}$, and has property $\mathbf{Z} \sim \mathcal{N}(\theta, \frac{\sigma^2}{N} I_k)$.

unbiasness: $\mathbb{E}[\mathbf{Z} | \mathbf{X}] = \frac{1}{N} \sum_{i=1}^N W_i W_i' \theta + \frac{\sigma}{N} \sum_{i=1}^N W_i \mathbb{E}[U_i | \mathbf{X}] = \theta$
variance:

$\mathbb{V}(Z_k | \mathbf{X}) = \mathbb{V}(\frac{1}{N} \sum_{i=1}^N \phi_k(X_i) Y_i | \mathbf{X}) = \frac{1}{N^2} \sum_{i=1}^N \phi_k(X_i)^2 \mathbb{V}(Y_i) = \frac{\sigma^2}{N}$

covariance:

$\mathbb{C}(Z_j, Z_k | \mathbf{X}) = \frac{\sigma^2}{N} \frac{1}{N} \sum_{i=1}^N \phi_j(X_i) \phi_k(X_i) = 0$

The Risk of MLE:

$\|\hat{m} - m\|^2 = \frac{1}{N} \sum_{i=1}^N (W_i' \hat{\theta}_{ML} - W_i' \theta)^2 = \frac{1}{N} \sum_{i=1}^N [W_i' (\hat{\theta}_{ML} - \theta)]^2$
 $= \frac{1}{N} \sum_{i=1}^N [\sum_{k=1}^K \phi_k(X_i) (Z_k - \theta_k)]^2 = \sum_{k=1}^K (Z_k - \theta_k)^2$

Loss function $L(Z, \theta) = \|Z - \theta\|^2 = \sum_{k=1}^K (Z_k - \theta_k)^2$
Risk: $R(Z, \theta) = \mathbb{E}[L(Z, \theta)] = \sum_{k=1}^K \mathbb{E}[(Z_k - \theta_k)^2] = \frac{K\sigma^2}{N}$
Has properties: $K \downarrow N \uparrow \sigma \downarrow \Rightarrow R \downarrow$
James-Stein Type Estimators:
Is the estimators: $\mathcal{L} = \{C_{K \times K} Z, C = \text{diag}\{c_1, \dots, c_K\}, c_k \in [0, 1]\}$

Risk: $\mathbb{E}[\sum_{k=1}^K (c_k Z_k - \theta_k)^2] = \underbrace{\frac{\sigma^2}{N} \sum_{k=1}^K c_k^2}_{\text{Var}(CZ)} + \underbrace{\sum_{k=1}^K (1 - c_k)^2 \theta_k^2}_{\text{Bias}^2}$

Oracle: $c_k^* = \frac{\theta_k^2}{\frac{\sigma^2}{N} + \theta_k^2}$, $\inf_{\hat{\theta} \in \mathcal{L}} R(\hat{\theta}, \theta) = \frac{\sigma^2}{N} \sum_{k=1}^K \left(\frac{\theta_k^2}{\frac{\sigma^2}{N} + \theta_k^2} \right) < \frac{K\sigma^2}{N}$

Stein's Unbiased Risk Estimate (SURE):

Consider an estimator $\hat{\theta}(Z)$, define g to be estimate minus the MLE:
 $g(Z) = \hat{\theta}(Z) - Z$

Risk: $\hat{R}_{\text{SURE}}(Z) = K\sigma^2 + 2\sigma^2 \sum_{k=1}^K \frac{\partial g_k(Z)}{\partial Z_k} + \sum_{k=1}^K (\hat{\theta}_k - Z_k)^2$

SURE g is weakly differentiable $\Rightarrow \mathbb{E}[\hat{R}_{\text{SURE}}(Z)] = \mathbb{E}[\|\hat{\theta} - \theta\|^2]$
The James-Stein Estimator

JS estimator: $\hat{\theta}_{\text{JS}}(Z) = \left(1 - \frac{(K-2)}{Z'Z} \frac{\sigma^2}{N}\right) Z$, $g(Z)_{\text{JS}} = -\left(\frac{(K-2)}{Z'Z} \frac{\sigma^2}{N}\right) Z$

With properties: $\sum_{k=1}^K \frac{\partial g_k(Z)}{\partial Z_k} = -\frac{(K-2)^2}{Z'Z} \frac{\sigma^2}{N}$,

$\hat{R}_{\text{SURE}}(Z) = \frac{K}{N} \sigma^2 - \frac{(K-2)^2}{Z'Z} \frac{\sigma^4}{N^2}$
 $R(\hat{\theta}_{\text{JS}}, \theta) = \frac{K}{N} \sigma^2 - (K-2)^2 \frac{\sigma^4}{N^2} \mathbb{E}\left[\frac{1}{Z'Z}\right] < \frac{K}{N} \sigma^2 = R(\hat{\theta}_{\text{ML}}, \theta), \quad K \geq 3$

Properties of Z : $Z_k^2 = \frac{\sigma^2}{N} \left(\frac{\theta_k}{\sigma/\sqrt{N}} + U \right)^2$, $U \sim N(0, 1)$, $Z_k \sim N(\theta_k, \frac{\sigma^2}{N}) \Rightarrow$

$Z'Z = \sum_{k=1}^K Z_k^2 \sim \frac{\sigma^2}{N} V$ where $V \sim \chi_{K+2W}^2$, $W \sim \text{Poisson}\left(\frac{p}{2}\right)$
for $K > 2$ and $S \sim \chi_K^2$, $\mathbb{E}\left[\frac{1}{S}\right] = \frac{1}{K-2}$

$\mathbb{E}\left[\frac{1}{Z'Z}\right] = \frac{N}{\sigma^2} \mathbb{E}\left[\frac{1}{V}\right] = \frac{N}{\sigma^2} \mathbb{E}\left[\mathbb{E}\left[\frac{1}{V} \middle| W\right]\right] = \frac{N}{\sigma^2} \mathbb{E}\left[\frac{1}{K-2+2W}\right]$
 $\geq \frac{N}{\sigma^2 (K-2+2\mathbb{E}[W])} = \frac{N}{\sigma^2 \left(K-2+N \sum_{k=1}^K \frac{\theta_k^2}{\sigma^2}\right)} = \frac{\sigma^2}{(K-2) \frac{\sigma^2}{N} + \|\theta\|^2}$

$R(\hat{\theta}_{\text{JS}}, \theta) \leq \frac{2}{N} \sigma^2 + \frac{(K-2) \frac{\sigma^2}{N} \|\theta\|^2}{(K-2) \frac{\sigma^2}{N} + \|\theta\|^2}$

Oracle vs. SURE
 $\hat{R}_{\text{SURE}}(Z, c) = \frac{K}{N} \sigma^2 - \frac{2}{N} \sigma^2 \sum_{k=1}^K (1 - c_k) + \sum_{k=1}^K Z_k^2 (1 - c_k)^2 =$
 $\frac{\sigma^2}{N} \sum_{k=1}^K c_k^2 + \sum_{k=1}^K \left(Z_k^2 - \frac{\sigma^2}{N} \right) (1 - c_k)^2$
variance reduction estimate of bias²

The JS shrinkage $\hat{c}_k = 1 - \frac{N-1}{Z_k^2} \frac{\sigma^2}{2}$ where $Z_k^2 = \frac{\theta^2}{N} + \theta_k^2 + \text{noise}$

Compared with Oracle shrinkage $c_k^* = \frac{\theta_k^2}{\frac{\sigma^2}{N} + \theta_k^2}$

In Practice

Set $Y_i = m(X_i) + \sigma U_i$, $M(X_i) = \sum_{k=1}^K \phi_k(X_i) \theta_k$, then

1. Use OLS to compute $\hat{\theta}_{MLE} = \mathbb{Z}$
2. Compute $\hat{c}_k = 1 - \frac{\frac{\sigma^2}{N}}{Z_k^2}$
3. $\hat{m}^{USE}(X_i) = \sum_{k=1}^K \phi_k(X_i) \hat{\theta}_k^{USE}$ (where $\hat{\theta}_k^{USE} = \hat{c}_k \hat{\theta}_{MLE} = \hat{c}_k \mathbf{Z}$)
 $\hat{m}^{OLS}(X_i) = \sum_{k=1}^K \phi_k(X_i) \hat{\theta}_k^{OLS}$

Bayesian Bootstrap

- A way to quantify uncertainty from a Bayesian perspective
- Introduction to simulation
- Bayesian bootstrap is also a weighted frequentist bootstrap

Setting

$\mathbf{Z} = (\underbrace{X}_{K \times 1}, \underbrace{Y}_{1 \times 1}) \in \{z_1, \dots, z_J\}$,

$\Pr(\mathbf{Z} = z_j | \underline{\theta} = \theta_j) \quad \text{where } \underline{\theta} = (\theta_1, \dots, \theta_J)'$,
 $\Theta = \mathbb{S}^{J-1} = \left\{ \underline{\theta} = (\theta_1, \dots, \theta_J)' \in \mathfrak{R}^3, \theta_j \geq 0, \sum_{j=1}^J \theta_j = 1 \right\}$

The data flow likelihood: $\Pr(\mathbf{Z} = z | \underline{\theta}) = \prod_{j=1}^J \theta_j^{N_j}$

Prior: Dirichlet distribution: $\pi(\theta_1, \dots, \theta_J) = \frac{\Gamma(\sum_{j=1}^J \alpha_j)}{\prod_{j=1}^J \Gamma(\alpha_j)} \left[\prod_{j=1}^J \theta_j^{\alpha_j - 1} \right]$

Gamma function, distribution

$\Gamma(x) = \int_{t=0}^{\infty} t^{x-1} \exp(-t) dt$
Properties: $\Gamma(x+1) = x\Gamma(x)$, $\Gamma(n) = (n-1)!$
Gamma Distribution: $f(x|\alpha, \beta) = \frac{\beta^\alpha x^{\alpha-1} e^{-\beta x}}{\Gamma(\alpha)}$, $x > 0, \alpha, \beta > 0$
 $E[X] = \alpha/\beta$ $var[X] = \alpha/\beta^2$

Reproductive Stable Property: if $X_j \sim \text{Gamma}(\alpha_j, \beta) \quad \forall j = 1, \dots, J$,
 $\sum_{j=1}^J X_i \sim \text{Gamma}(\sum_{j=1}^J \alpha_j, \beta)$
 $\text{Gamma}(1, \beta) = \text{Exp}(\beta)$, $\Gamma(u/2, 2) = \chi_u^2$

Relationship to Dirichlet distribution: Let $\{W_j\}_{j=1}^J$ be independent random variables with $W_j \sim \text{Gamma}(\alpha_j, 1)$, $V_j = \frac{W_j}{\sum_{i=1}^J W_i}$, let $\sum_j V_j = 1$

then $(V_1, \dots, V_J)$ follows Dirichlet distribution with parameter $\alpha = (\alpha_1, \dots, \alpha_J)'$.
Posterior Distribution

Given $\Pr(\mathbf{Z} = z | \underline{\theta}) = \prod_{j=1}^J \theta_j^{N_j}$ we have likelihood density
 $f(z | \underline{\theta}) = \frac{N!}{N_1! \dots N_J!} \prod_{j=1}^J \theta_j^{N_j}$

Using Bayes' rule, posterior $\pi(\underline{\theta} | z) = \frac{\Gamma(\sum_{j=1}^J N_j + \alpha_j)}{\prod_{j=1}^J \Gamma(N_j + \alpha_j)} \prod \theta_j^{N_j + \alpha_j - 1}$,

$\alpha_j = N_j + \alpha_j$.

Improper Prior

$F_N^{\text{BB}}(z) = \sum_{i=1}^N V_i^* \mathbf{1}(Z_i \leq z) = \sum_{i=1}^N V_i^* \sum_{j=1}^J \mathbf{1}(Z_i = z_j) \mathbf{1}(z_j \leq z)$
 $= \sum_{j=1}^J \mathbf{1}(N_j > 0) \left(\sum_{i=1}^N \mathbf{1}(Z_i = z_j) V_i^* \right) \mathbf{1}(z_j \leq z) = \sum_{j=1}^J V_j \mathbf{1}(z_j \leq z) = F^{\text{BB}}(z)$
Application to Linear Regression

- (1) Draw a vector of N iid $\textit{Gamma}(1,1)$, sum normalized to $V_{(s)}^\#$
- (2) $\beta^{(s)} = \left[\sum_{j=1}^N V_{(s),j}^\# X_j X_j^T \right]^{-1} \left[\sum_{j=1}^N V_{(s),j}^\# X_j Y_j \right]$
- (3)Collect the resulting estimates $\beta^{(1)}, \beta^{(2)}, \dots, \beta^{(S)}$ to form the posterior distribution, yield confidence interval.

Long, Short, Residual Regression

Settings: $(W, X, Y) \overset{i.i.d.}{F}$
Long regression: $\mathbb{E}^*[Y|X, W] = X^l \beta_0 + W^l \gamma_0$

Short regression: $\mathbb{E}^*[Y|X] = X^l b_0$

Auxiliary Regression: $\mathbb{E}^*[W|X] = \underbrace{\Pi_0}_{J \times 1} \underbrace{X}_{J \times K}^{K \times 1}$

Property: $\beta_{0,LR} = b_{0,SR} - \Pi_0^l \gamma_0$
Law of Iterated Linear Predictors (LILP): $\mathbb{E}^*[Y|X] = \mathbb{E}^*[\mathbb{E}^*[Y|X, W]|X]$
Residual Regression: settings $V = W - \mathbb{E}^*[W|X]$

Hajek Projection Property:
If $\text{Cov}(X, W) = 0$, then: $\mathbb{E}^*[Y|(X, W)] = \mathbb{E}^*[Y|X] + \mathbb{E}^*[Y|W] - \mathbb{E}[Y]$
If $X_i \perp\!\!\!\perp X_j$, then $\mathbb{E}^*[Y|X_1, \dots, X_K] = \sum_{j=1}^K \mathbb{E}^*[Y|X_j] - (K-1)\mathbb{E}Y$
Frisch-Waugh Theorem:
 $\mathbb{E}^*[Y|(X, V)] = \mathbb{E}^*[Y|X] + \mathbb{E}^*[Y|V] - \mathbb{E}[Y]$,
 $\mathbb{E}^*[Y|(X, V)] = \mathbb{E}^*[Y|(X, W)] = Xb_0 + W\gamma_0 = Xb_0 + V\gamma_0$
Useful facts: $\mathbb{E}^*[Y|X] = \mathbb{E}[Y] + \mathbb{C}(X, Y')\mathbb{V}(X)^{-1}[X - \mathbb{E}[X]]$

Finite and Large Sample Properties of OLS

Assumptions: $(X, Y) \sim F_{X,Y}$
A.1 Regularity condition:
 $\mathbb{E}\left[Y^2\right] < \infty$, $\mathbb{E}\left[\|X\|^2\right] < \infty$, $\mathbb{E}\left[XX^T\right]$ is positive definite
A.1 implies that $\beta_0 = \mathbb{E}\left[XX^T\right]^{-1}\mathbb{E}[XY]$ exists and unique MSE.

A.2 Random Sampling: $(X_i, Y_i) \overset{i.i.d}{F_{X,Y}}$

A.2 implies $\hat{\beta} = \left(X^T X\right)^{-1} X^T Y$

A.3 Non-singular $X^T X$: equals to $D = \det(X^T X) \neq 0$
 $\mathbb{E}[Y \mid X = x] = x^T \beta_0 \Rightarrow \mathbb{E}[b(X, Y) \mid X = x] = \left(X^T X\right)^{-1} X^T X \beta_0 = \beta_0$

finite sample properties:
 $\mathbb{E}(b(X, Y) \mid X, D \neq 0) = \mathbb{E}[b(X, Y) \mid D \neq 0] = \beta_0$
If $\mathbb{E}[\|b(X, Y)\|] < \infty$, then $\mathbb{E}[b(X, Y)] = \beta_0$
Variance: $\mathbb{V}[b(X, Y) \mid X = x] = (X^T X)^{-1} X^T \mathbb{V}[Y \mid X = x] X (X^T X)^{-1}$
Formula: given $A = (X^T X)^{-1} X^T$, $\mathbb{V}(AY) = A \mathbb{V}(Y) A^T$
 $\Sigma(X) = \mathbb{V}[Y|X = x] = \text{diag}\left(\sigma^2(X_1), \dots, \sigma^2(X_N)\right)$, $\sigma^2(X_i) = \mathbb{V}(Y_i|X_i = x_i)$
 $\mathbb{V}[b(X, Y)|X = x] = \left(\sum_{i=1}^N X_i X_i^T\right)^{-1} \times \left(\sum_{i=1}^N X_i X_i^T \sigma^2(X_i)\right) \times \left(\sum_{i=1}^N X_i X_i^T\right)^{-1}$
Assuming homoscedasticity: $\Sigma(X) = \sigma^2 \sum_{i=1}^N X_i X_i^T$
Summary:
 $\mathbb{E}[b(D \neq \hat{0})] = \beta_0$
 $\mathbb{V}[b|X = x] = \left(\sum_{i=1}^N X_i X_i^T\right)^{-1} \times \left(\sum_{i=1}^N \sigma^2(X_i) X_i X_i^T\right) \times \left(\sum_{i=1}^N X_i X_i^T\right)^{-1}$

Asymptotic Sampling Distribution

Convergence in probability:
 $\lim_{N \rightarrow \infty} \Pr\left(|Q_N - \alpha| > \varepsilon\right) = 0 \quad \forall \varepsilon > 0 \Rightarrow Q_N \xrightarrow{P} \alpha$
LLN: $\frac{1}{N} \sum_{i=1}^N Y_i \xrightarrow{P} \mathbb{E}[Y_i]$
Slutsky's Theorem: If (i) a sequence of random variables Q_N takes on values in $\mathbb{R}^J$, (ii) $Q_N \xrightarrow{P} \alpha$ and (iii) the function $g: \mathbb{R}^J \rightarrow \mathbb{R}^M$ is continuous at α , then $g(Q_N) \xrightarrow{P} g(\alpha)$

Property of $\hat{\beta}$: $\hat{\beta} \xrightarrow{P} \mathbb{E}[X_i X_i^T]^{-1} \mathbb{E}[X_i Y_i] = \beta_0$
Convergence in Distribution of $\hat{\beta}$:
Let $U_i = Y_i - X_i^T \beta_0$. Then
 $\hat{\beta} = \textit{beta}_0 + \left(\frac{1}{N} \sum_{i=1}^N X_i X_i^T\right)^{-1} \times \left(\frac{1}{N} \sum_{i=1}^N X_i U_i\right) = \underbrace{\beta_0}_{\text{truth}} + o_p(1)$

$\sqrt{N}(\hat{\beta} - \beta_0) = \left(\frac{1}{N} \sum_{i=1}^N X_i X_i^T\right)^{-1} \left(\frac{1}{\sqrt{N}} \sum_{i=1}^N X_i U_i\right)$
 $\mathbb{V}\left(\frac{1}{\sqrt{N}} \sum_{i=1}^N X_i U_i\right) = \frac{1}{N} \sum_{i=1}^N \mathbb{V}(X_i U_i) = \mathbb{V}(X_i U_i)$
CLT: $\frac{1}{\sqrt{N}} \sum_{i=1}^N Y_i \xrightarrow{d} \mathcal{N}(0, \Sigma)$
Slutsky's Theorem: Let S_N be a sequence of $K \times 1$ random variables with $S_N \xrightarrow{d} \mathcal{N}(0, \Sigma)$ and let Q_N be a $J \times K$ random variable with $Q_N \xrightarrow{P} \alpha$. Then: $Q_N \cdot S_N \xrightarrow{d} \mathcal{N}\left(0, \alpha \Sigma \alpha^T\right)$
1. Let $\mathbb{E}[X_i X_i^T]^{-1} = \Gamma_0^{-1}$

2. Let $\mathbb{E}[X_i U_i U_i^T X_i^T] = \mathbb{E}[X_i X_i^T U_i^2] = \Omega_0$

$\sqrt{N}(\hat{\beta} - \beta_0) \xrightarrow{d} \mathcal{N}(0, \Gamma_0^{-1} \Omega_0 \Gamma_0^{-1 T})$, $\hat{\Gamma}^{-1} \hat{\Omega} \hat{\Gamma}^{-1 T} \xrightarrow{P} \Gamma_0^{-1} \Omega_0 \Gamma_0^{-1 T}$
 $\hat{\beta} \overset{\text{approx}}{\sim} \mathcal{N}\left(\beta_0, \frac{\Gamma_0^{-1} \Omega_0 \Gamma_0^{-1 T}}{N}\right)$

Testing and confidence intervals
 $\sqrt{N}\left(\hat{\beta}_k - \beta_k\right) \xrightarrow{d} \mathcal{N}(0, \Lambda_{kk})$, $T_N = \sqrt{N} \Lambda_{kk}^{-\frac{1}{2}}\left(\hat{\beta}_k - \beta_k\right) \xrightarrow{d} \mathcal{N}(0, 1)$
Then 95% interval $\left(\hat{\beta}_k - 1.96 \frac{\Lambda_{kk}}{\sqrt{N}} \leq \beta_k \leq \hat{\beta}_k + 1.96 \frac{\Lambda_{kk}}{\sqrt{N}}\right)$

Distribution

Distribution	PDF	MGF
Bernoulli	$P(X = 1) = p$	$1 - p + pe^t$
Geom	$(1 - p)^{k-1} p$	$\frac{pe^t}{1 - (1 - p)e^t}$
Bin	$\binom{n}{k} p^k q^{n-k}$	$\left(1 - p + pe^t\right)^n$
NBin	$k \mapsto \binom{k+r-1}{k} \cdot (1 - p)^k p^r$	$\left(\frac{p}{1 - e^t + pe^t}\right)^r$
Poisson	$\frac{\lambda^k e^{-\lambda}}{k!}$	$e^{\lambda(e^t - 1)}$
Uniform	$\text{U}(a, b)$	$\frac{e^{tb} - e^{ta}}{t(b - a)}$
Normal	$\frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$	$e^{t\mu + \frac{1}{2}\sigma^2 t^2}$
Gamma	$f(x) = \frac{1}{\Gamma(k)\theta^k} x^{k-1} e^{-x/\theta}$	$(1 - \theta t)^{-k}$ for $t < \frac{1}{\theta}$
Beta	$\frac{x^{\alpha-1}(1-x)^{\beta-1}}{\frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}}$	$1 + \sum_{k=1}^{\infty} \left(\prod_{r=0}^{k-1} \frac{\alpha+r}{\alpha+\beta+r}\right) \frac{t^k}{k!}$
Exponential	$\lambda e^{-\lambda x}$	$\left(1 - \frac{t}{\lambda}\right)^{-1}$
Multi-Norm	$N(\mu, \Sigma)$	$e^{t^T(\mu + \frac{1}{2}\Sigma t)}$
Chi2	$\frac{x^{k/2-1} e^{-x/2}}{2^{k/2}\Gamma(k/2)}$	$(1 - 2t)^{-k/2}$

Distribution	E	Var		
Bernoulli	$\frac{p}{p}$	$\frac{p(1-p)}{p(1-p)}$		
Bin	np	$\frac{np(1-p)}{(1-p)}$	Dist.	CF
Geom	$\frac{1-p}{p}$	$\frac{p^2}{p^2}$	Gamma	$(1 - \theta it)^{-k}$
Exponential	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$	Multi-Normal	$\exp\left(i\mu^T t - \frac{1}{2}t^T \Sigma t\right)$
Uniform	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$	Poisson	$\exp\left[\lambda\left(e^t - 1\right)\right]$
Poisson	λ	λ	Exp	$\frac{\lambda}{\lambda - it}$
Chi2	k	$2k$		
Gamma	$k\theta$	$k\theta^2$		
Beta	$\frac{\alpha}{\alpha+\beta}$	$\frac{\alpha-1}{\alpha+\beta-2}$		

Useful Facts

Characteristic function:
(1) $X \overset{d}{=} Y \Leftrightarrow \psi_X = \psi_Y$ (2) $X \perp\!\!\!\perp Y \Leftrightarrow \psi_{X+Y} = \psi_X \psi_Y$
Cauchy-Schwarz Ineq: $\mathbb{E}[|X'Y|] \leq (\mathbb{E}[|X|^2])^{\frac{1}{2}} (\mathbb{E}[|Y|^2])^{\frac{1}{2}}$
Some Useful properties of Regression

Consider $\mathbb{E}^*[Y|(1, X')]$. Then $\sqrt{T}\left[\left(\hat{\hat{\theta}}\right) - \left(\frac{\alpha_0}{\theta_0}\right)\right] \xrightarrow{d} \mathcal{N}(0, \Omega)$

Where $\Omega = \sigma^2 \mathbb{E}\left(\left(\frac{1}{X}\right) \left(1 \ X'\right)^{-1} = \sigma^2 \left(\begin{smallmatrix} 1 & \mathbb{E}[X'] \\ \mathbb{E}[X] & \mathbb{E}[XX'] \end{smallmatrix}\right)^{-1}\right)$

If $\sqrt{T}\left[\left(\hat{\hat{\theta}}\right) - \left(\frac{\alpha_0}{\theta_0}\right)\right] \xrightarrow{d} \mathcal{N}(0, \Omega)$ where $\Omega = \left(\begin{smallmatrix} \Omega_{11} & \Omega_{12} \\ \Omega_{21} & \Omega_{22} \end{smallmatrix}\right)$

Then $\sqrt{T}[\hat{\theta} - \theta_0] \xrightarrow{d} \mathcal{N}(0, \Omega_{22})$
Block inverse mat:
 $\begin{pmatrix} A & B \\ C & D \end{pmatrix}^{-1} = \begin{pmatrix} (A - BD^{-1}C)^{-1} & \\ & (D - CA^{-1}B)^{-1} \end{pmatrix} \begin{pmatrix} I & -BD^{-1} \\ -CA^{-1} & I \end{pmatrix}$

Then $\Omega_{22} = \sigma^2[\mathbb{E}X X' - \mathbb{E}X \mathbb{E}X']^{-1} = \sigma^2 \mathbb{V}(X)^{-1}$
Delta Methods: Consider $\sqrt{T}[\hat{\theta} - \theta_0] \xrightarrow{d} \mathcal{N}(0, \Lambda)$
Then $\sqrt{T}[e^l \hat{\theta} - e^l \theta_0] \xrightarrow{d} \mathcal{N}(0, e^l \Lambda e)$, $\frac{\sqrt{T}[e^l \hat{\theta} - e^l \theta_0]}{\sqrt{e^l \Lambda e}} \xrightarrow{d} \mathcal{N}(0, 1)$
and $\frac{T[e^l \hat{\theta} - e^l \theta_0]^2}{e^l \Lambda e} \xrightarrow{d} \chi_1^2$
Moments: Skewness: $\frac{\mu_3}{\mu_2^{3/2}}$, Kurtosis: $\frac{\mu_4}{\mu_2^2}$

Conditional moment:
For all $g(\cdot) \in \mathcal{G}$, $E[Y|X] = \arg \min_{g(\cdot) \in \mathcal{G}} MSE_Y[g(X)]$

Inequalities:

- Markov: $P[g(X) \geq a] \leq \frac{E[g(X)]}{a}$ for any $a > 0$
- Chebyshev: $P[|X - \mu| \geq \varepsilon] \leq \frac{\sigma^2}{\varepsilon^2}$
- Jensen: $g(E(X)) \leq E(g(X))$
- Young: $\frac{1}{p} + \frac{1}{q} = 1$, $ab \leq \frac{1^p}{p} + \frac{b^q}{q}$
- Norm: $\|X\|_p = (E(|X|^p))^{\frac{1}{p}}$
- Holder: $\frac{1}{p} + \frac{1}{q} = 1$, $\|XY\|_1 = E[|XY|] \leq \|X\|_p \|Y\|_q$
- Covariance: $[Cov(X, Y)]^2 \leq Var(X)Var(Y)$

Order statistics: Suppose $X_i \sim U[0, \theta]$, then $E(X)_{(N)} = \frac{N}{N+1} \theta$, $Var(X)_{(N)} = \frac{N}{(N+2)(N+1)} \theta^2$

Gaussian integral: $\int_{-\infty}^{+\infty} e^{-ax^2+bx+c} dx = \sqrt{\frac{\pi}{a}} e^{\frac{b^2}{4a}+c}$
Constants:
97.5 quantile of $\mathcal{N}(0, 1)$: 1.96
95 quantile of χ_1^2 : 3.84